SMART INDIA HACKATHON 2026

Case Study: ApnaGig

Cooperative Gig Services Platform for Household & Community Services

Round 2 Deep-Dive: Anticipated Judge Questions, Operational Answers & Competitive Rationale

Problem Statement ID: 26089
Theme: Agriculture, FoodTech & Rural Development
PS Category: Software
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1. Purpose of This Case Study

This document consolidates the operational, financial, legal, and technical edge cases that judges are likely to probe in the Round 2 evaluation of ApnaGig. Each question is answered against the systems we have already designed — the Job Lifecycle Service, Grievance & Accountability System, AI Smart Automation Engine, and Cooperative-backed verification layer — so that every answer demonstrates depth already built into the architecture, not improvisation.

The questions are grouped into four categories that mirror how a judging panel typically probes a gig-economy pitch: Trust & Dispute Resolution, Worker & Cooperative Fairness, Business Model & Sustainability, and Technology, Access & Governance. A final section makes the explicit case for why ApnaGig's cooperative-first approach is structurally better suited to rural India than a transplanted urban gig-app model.

2. Trust, Accountability & Dispute Resolution

Q: If a customer hires a group of labourers and refuses to pay after the work is completed, what happens?

A: Payment is pre-authorized at booking (UPI mandate or card hold) and only captured after the worker marks completion via a job-end OTP and a grace-period dispute window (2–4 hours) passes without objection. If the customer disputes in bad faith, the case goes to the Grievance & Accountability System, where a cooperative official reviews GPS check-in/out logs, timestamped photos, and the OTP trail. For verified genuine work, the worker can be paid from the welfare fund immediately while the platform separately recovers dues from the customer — protecting the worker's cash flow rather than making them wait on a dispute. Repeat non-paying customers are flagged and required to fully prepay for all future bookings.

Q: If the work finishes earlier than the agreed timeline, for example 8 days instead of 10 — is the worker paid in full?

A: Yes. Multi-day jobs are priced against the agreed scope of work, not the number of days consumed, so finishing early does not reduce the payout. Completion is verified through a task checklist plus photo/video evidence and customer sign-off in the Job Lifecycle Service, which distinguishes 'finished early' from 'incomplete.' Workers who consistently deliver ahead of schedule get a rating boost, which the AI matching engine uses to prioritize them for higher-value future jobs — so efficiency is rewarded rather than penalized, which is the opposite incentive structure of hourly-billed platforms.

Q: If a labourer damages or spoils the work, who is held responsible?

A: Liability sits with a cooperative-backed insurance/liability fund, not the individual worker personally, unless negligence is proven. The Incident Service (already part of our architecture) logs photo/video evidence and metadata for the case. The cooperative arbitrates: for minor damage the worker receives a rating deduction and a formal warning; for major or repeated damage, the liability fund compensates the customer directly and the worker is routed into the Skill Gap Detection & Training Recommendation module for mandatory retraining before being matched again. This keeps the customer whole without leaving an individual worker financially exposed for an honest mistake, while still creating a real consequence for negligence.

Q: What if a worker is injured while on the job — who bears responsibility?

A: Every verified worker is covered under the Insurance/Welfare API integration shown in our technical architecture, which links to government and cooperative-backed insurance schemes. An on-site injury is logged through the Incident Service with photo/video evidence and an SOS trigger if urgent, which routes directly to the Emergency SOS & Live Tracking feature for immediate help. The claim is then processed through the linked insurance scheme rather than becoming a liability dispute between customer and worker — this is one of the concrete reasons the welfare layer is integrated into the core platform rather than being an optional add-on.

Q: What if the dispute is about subjective quality — e.g. 'the cleaning wasn't good enough' — rather than physical damage?

A: Subjective quality disputes are handled differently from damage claims: the Grievance & Accountability module supports a structured complaint with before/after photo evidence (many service categories in our Workers App already capture job-completion photos as part of the closing flow). If evidence is inconclusive, the cooperative can offer a partial goodwill credit or a follow-up visit rather than a full refund, preserving the worker's earned pay while still protecting customer trust. Recurring quality complaints against the same worker feed into their skill-recognition score and can trigger a recommended retraining module, rather than an automatic punitive deduction — since subjective disputes shouldn't be treated with the same certainty as documented damage.

Q: If multiple workers are assigned to one job, how is payment split, and who is accountable if the group underperforms?

A: Group jobs are booked against a single job record in the Booking Service, with each worker's individual contribution (role, hours, task subset) logged separately in the Job Lifecycle Service — so payment splits are proportional and auditable, not an even split by default. If the group underperforms, the Incident Service and Grievance system attribute responsibility at the individual level using task-level completion logs, rather than penalizing the whole group collectively, so one worker's failure doesn't wipe out another's fair earnings.

3. Worker & Cooperative Fairness

Q: What happens if a worker accepts a job and then simply doesn't show up?

A: The Job Lifecycle Service tracks an accept-to-start OTP window. If the worker doesn't check in within that window, the job is automatically re-routed to the next available verified worker through the AI matching engine, minimizing customer wait time. The no-show is recorded against the worker's reliability score, and repeated no-shows deprioritize that worker in future AI-driven job allocation — a self-correcting mechanism rather than a manual disciplinary process.

Q: How do you prevent fake ratings or collusion between a customer and a worker?

A: The Fraud Detection Service, already part of our microservices layer, flags anomaly patterns such as the same customer-worker pair repeatedly transacting outside typical geographic or pricing norms, GPS spoofing on check-in/out, or rating patterns inconsistent with actual job completion data. Flagged cases are routed to cooperative officials for manual review before any rating or payout is affected, so the system assists human oversight rather than fully automating punitive action.

Q: The whole pitch relies on cooperatives being fair — what stops a cooperative's own leadership from exploiting the workers it represents?

A: Two structural checks: first, the Digital Cooperative Identity and Federation Intelligence Dashboard make every worker's registration, job history, and earnings visible and portable — a worker isn't locked into a single cooperative's opaque records the way informal labour arrangements are today. Second, welfare fund contributions and payouts are logged transparently and viewable in-app by the worker, not just held by cooperative officials, so misuse is visible rather than hidden in a paper ledger. This doesn't eliminate the need for cooperative governance reform, but it removes the specific problem of invisible bookkeeping, which is the main lever used to exploit workers in unorganized cooperatives today.

Q: Are the gig workers on this platform treated as employees or independent contractors, and how do you handle labour law compliance?

A: Workers remain independent contractors affiliated with their registered cooperative — consistent with how cooperative labour arrangements already operate under India's cooperative societies laws — rather than being platform employees, which avoids ambiguous employment-classification liability for us. Compliance obligations (minimum wage benchmarks, welfare contributions, insurance) are handled at the cooperative level, which is already the legally recognized entity, while our platform provides the digital infrastructure layer on top. This is deliberately different from urban gig-platform employment disputes, because the cooperative — not the app — is the legal employer-equivalent relationship.

4. Business Model & Sustainability

Q: What is your plan to validate product-market fit before scaling statewide or nationally?

A: We propose an MVP rollout in a small number of partner cooperatives in a single district, focusing on 2–3 service categories (e.g. farm labour and household services) to validate booking-to-completion conversion, payment reliability, and worker retention before expanding service categories or geography. Success metrics would include repeat booking rate, dispute rate per 100 jobs, and worker earnings uplift versus their prior informal arrangement — concrete, measurable signals rather than raw download counts, which is more relevant to a platform whose value proposition is trust and fairness, not just reach.

Q: Agricultural and household service demand is seasonal — how do you avoid overloading workers during peak periods like harvest season?

A: The Monthly Capacity Forecasting and Demand Prediction (Geo & Seasonal) modules in our AI Smart Automation Engine are explicitly designed to anticipate seasonal spikes ahead of time, allowing cooperatives to pre-recruit or reallocate workers from lower-demand categories in advance rather than reactively overloading existing verified workers. The Fair

Work Allocation Algorithm also caps job assignment per worker based on workload thresholds, so demand spikes are absorbed by spreading jobs across a wider verified pool rather than over-assigning to the most available individuals.

5. Technology, Access & Governance

Q: What stops a customer from booking through the app once, then going direct to the same worker to avoid the platform going forward?

A: The value the platform adds — insurance coverage, a digital invoice/paper trail, access to the grievance and welfare system, and verified worker replacement if the original worker is unavailable — only applies to bookings made through ApnaGig. Off-platform arrangements lose all of these protections for both sides, which is the retention mechanism: unlike a pure marketplace where the app is just a discovery layer, our value is embedded in the ongoing transaction, not just the introduction.

Q: Why would workers trust a digital welfare micro-contribution over simply receiving full cash-in-hand?

A: Because the contribution, balance, and payout history are fully visible to the worker in-app in real time, unlike existing informal welfare schemes where contributions often disappear into an opaque cooperative ledger. Transparency — not the mechanism of deduction itself — is what is designed to build trust here, and the contribution rate is configurable so cooperatives can tune it to what their specific worker base is comfortable with rather than imposing a fixed national rate.

6. Why ApnaGig Is the Optimal Solution

Existing consumer gig platforms (UrbanClap, Uber/Uber Eats, TaskRabbit and similar) were built for dense urban markets and treat the worker as an interchangeable supply-side resource. Rural India's problem is structurally different: workforce trust, seasonal/agricultural demand patterns, and pre-existing cooperative social structures are all first-class constraints, not edge cases. ApnaGig is designed around that difference rather than adapting an urban template.

6.1 Feature-Level Comparison

Feature / Parameter	UrbanClap, Uber & Uber Eats, TaskRabbit, etc.	ApnaGig
Demand Forecasting	No	Yes
Workforce Planning	No	Yes
Fair Work Allocation	No	Yes
Worker Ownership / Cooperative Model	No	Yes
Worker Welfare	No	Yes
Incident Evidence	No	Yes
Community Services	No	Yes
Federation Administration	No	Yes
Fraud Detection	No	Yes

6.2 Answering the 'Why not just use an existing app' Objection

Urban gig platforms could technically expand into rural service categories, but they would still operate on a commission-extraction model, lack integrated worker welfare, and have no mechanism to plug into India's existing cooperative and government infrastructure (e-Shram, ICAR, National Cooperative Database). ApnaGig's core differentiation is not a single feature — it is that the entire system (ownership model, AI allocation, and welfare) is built around strengthening the cooperative structures that already exist in rural India, rather than disintermediating them the way urban platforms disintermediate individual contractors.

6.3 Summary Positioning Statement

> *ApnaGig is not a rural clone of an urban gig app — it is a digital operating layer for India's existing rural cooperative economy, which is precisely why the trust, welfare, and fairness mechanisms are built into the architecture rather than bolted on afterward.*

ApnaGig — Fee & Welfare Contribution Structure

Platform Fee, Welfare Amount & Charges — Customer Side vs Worker Side (Worked Example)

1. Customer Side

The customer pays the base job value plus the platform/cooperative service fee plus the payment gateway charge. The customer never pays into the welfare fund directly — that is entirely on the worker side.

Component	% of Job Value	Amount (₹1,000 job)
Base job value	100%	₹1,000
Platform / cooperative service fee	+5%	+₹50
Payment gateway charge (pass-through)	+2%	+₹20
Total customer pays	107%	₹1,070

From the customer's perspective, this is a flat 7% service charge on top of the job value — comparable to a standard booking or convenience fee on any service marketplace.

2. Worker Side

The worker receives the base job value minus a self-directed welfare contribution. The worker never pays the platform fee or the payment gateway charge — those are entirely on the customer side.

Component	% of Job Value	Amount (₹1,000 job)
Base job value	100%	₹1,000
Welfare contribution (self-directed, into own insurance/training fund)	−3%	−₹30
Worker take-home	97%	₹970

The 3% welfare contribution is visible in-app to the worker as their own running balance (insurance, skill training fund) — not a platform cut.

3. Combined Breakdown (₹1,000 Job Example)

Component	Who Pays	Amount (₹1,000 job)
Worker earnings (take-home)	—	₹970
Welfare contribution	Deducted from worker's share	₹30
Platform / cooperative service fee	Charged to customer	₹50
Payment gateway charge	Charged to customer (pass-through)	₹20